

Display entity chained interpolation has inconsistent behavior

MC-260874 Entities

Status	Resolved / Fixed
Affected	1.19.4 Release Candidate 1, 1.19.4
Fixed in	23w13a
Reported	2023-03-09

STEPS TO REPRODUCE

1. Create a void superflat world.
2. Run `/scoreboard objectives add test dummy``.
3. Place a repeating command block followed by chain command blocks, and enter these commands in order:
``execute positioned 0 -58 0 run summon minecraft:block_display ~ ~ ~ {Tags:[bob],block_state:{Name:"stone"},transformation:{translation:[0.0f,0.0f,0.0f],scale:[0.0f,0.0f,0.0f],left_rotation:[0.0f,0.0f,0.0f,1.0f],right_rotation:[0.0f,0.0f,0.0f,1.0f]},start_interpolation:0,interpolation_duration:0}``
``scoreboard players add @e[tag=bob] test 1``
``execute as @e[tag=bob,scores={test=2}] run data merge entity @s {transformation:{scale:[0.1f,0.1f,0.1f],translation:[0.0f,2.0f,0.0f]},start_interpolation:0,interpolation_duration:0}``
``execute as @e[tag=bob,scores={test=20}] run data merge entity @s {transformation:{scale:[0.1f,0.1f,0.1f],translation:[0.0f,0.0f,0.0f]},start_interpolation:0,interpolation_duration:40}``
``kill @e[tag=bob,type=minecraft:block_display,scores={test=50..}]``
4. Wait until the command chain reaches the kill condition at a score of 50 or higher.
5. Observe the block display's animation and compare it with the expected single animation from scale ``0.1`` and translation ``0,2,0`` to translation ``0,0,0`` over 40 ticks.
6. Pause the game for different lengths of time, then allow the command chain to continue.
7. Observe that the same commands can produce different animations: the block display may expand from scale 0 at translation ``0,0,0``, expand from an intermediate scale and translation, or remain at scale ``0.1`` while moving from translation ``0,2,0`` to ``0,0,0``.

OBSERVED BEHAVIOUR

When multiple transformation updates are applied to a display entity in sequence, the entity does not always begin the next interpolation from the transformation established by the preceding command. In affected versions, a later interpolation can reuse stale state from an earlier transformation and apply the newest interpolation duration to it, even when the earlier update used an interpolation duration of zero or had already completed. This can cause an instantaneous, invisible intermediate animation before the next visible animation begins, leaving the entity at an incorrect position or scale. For example, an entity that is set immediately to a scale of 0.1 and a translation of 0,2,0 can later begin its 40-tick interpolation from a partially interpolated position or scale instead of from that exact transformation. The resulting animation may therefore appear to start at an unexpected height, with an unexpected scale, or from a point partway through a previous animation. This animation-leaking behavior was fixed in 23w13a, but deterministic chained interpolation commands could still produce different visible animations after pausing or under different tick-update timing because of the separate desynchronization issue tracked as MC-261600.

EXPECTED BEHAVIOUR

Display entities should apply chained transformation updates consistently and in order. An instantaneous transformation should take effect immediately, without replaying or partially reapplying an earlier interpolation, and a subsequent animation should begin from that exact current transformation and reach its target over the specified duration. The same commands should always produce one deterministic animation, regardless of command timing or pausing.

ENVIRONMENT

Minecraft Java Edition Versions: 1.19.4 Release Candidate 1, 1.19.4

ORIGINAL DESCRIPTION

Chained interpolation animations appear to have inconsistent behaviors.

This bug was partially fixed in 23w13a. The issue with Animation leaking is resolved, but the issue with Multiple animations from desync still happens. I have separated the second bug into

MC-261600

Resolved

so that this one can be treated separately. I have kept the original description and repros here for context.

From my understanding, this is a result of two different bugs happening together that may affect a large amount of animations.

Bug 1: Animation leaking

Mid-animation interpolation is fundamentally broken, as it is prone to leaking and ghost-animating the previous animation instantaneously using the duration of the current animation, before starting the next animation, even if the previous animation is already finished.

To go more in depth: The following animation may continue at a position partway along the previous animation, even if the previous animation finished and even if the previous animation should have been instant with 0 duration, indicating that it somehow assumed there was an animation when there was none, by applying the duration of the latest animation to the previous animation. Additionally, it may continue from where the previous animation currently is mid-animation, but from the wrong point in time, and hence not continuing from the proper location.

Bug 2: Multiple animations from desync

See separate bug:

MC-261600

Resolved

The game can desync, for example, depending on how much the user pauses. This then becomes evident when a transform animation is applied, which can cause animations to behave in multiple different ways based on how desynced they are.

Animations may appear to manifest in multiple ways with a single set of deterministic commands that are different from each other when desynced, when the expected behavior is a single animation.

Pausing the game for various lengths of time may trigger this desync. See: desyncpause.mp4 and desyncpause2.mp4. This desync appears to be due to interpolation occasionally happening in the tick after it was set, resulting in different forms of the animation appearing for differe

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-260874>